Behavioural Research in Statistical Methods - Project Report - 1

Experiment : Sentence Memorability

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1. Introduction :

The present experiment is on memorability in sentences. According to the research, the paper wants to experiment how humans memorize sentences. And how it is dependent on the words used in the sentence and their memorability and also framing those words. Users have to see sentences randomly displayed on screen and guess whether it is repeated. High-memorability words tend to be concrete, imageable, and emotionally rich, while low-memorability words tend to be abstract and semantically impoverished (Rubin & Friendly, 1986; Shepard, 1967). This hierarchy of memorability, originally documented at the level of individual words and pictures, raises an important question about how memory operates at the level of full sentences.

The main aim of this experiment is to find how sentence memorability varies as a function of constituent word memorability and sentence voice? Preliminary analyses establish whether the observed condition differences are statistically reliable and whether voice is a meaningful source of variation.

1.1 Hypothesis :

Statement-1 : This tests the effect of the noun types (LL vs. HH/HL/LH) on the sentence's memorability. H0(Null): There is no significant difference in corrected memorability scores between sentences in the LL condition and sentences in the HH, HL, or LH conditions.Ha (Alternative): Sentences in the LL condition will have significantly lower corrected memorability scores than sentences in the HH, HL, or LH conditions.

Statement-2 : This tests the effect of the grammatical structure (Active vs. Passive) on memorability. H0 (Null): There is no significant difference in corrected memorability scores between passive-voice sentences and active-voice sentences. Ha (Alternative): Passive-voice sentences will yield significantly lower corrected memorability scores compared to active-voice sentences.

2. Method Overview

2.1 Experimental Design

This study used a continuous recognition paradigm. Participants viewed sentences and indicated whether each sentence was new or previously presented. The Participants have to interrupt whenever they see a sentence which is already seen by clicking the spacebar (basically they generate an event). Sentences were constructed using a Subject–Verb–Object (S–V–O) format and varied systematically according to:

Word Memorability (4 levels):

- HH (High subject, High object)
- HL (High subject, Low object)
- LH (Low subject, High object)
- LL (Low subject, Low object)

Voice (2 levels):

- Active
- Passive

This resulted in 8 experimental conditions.

High and low memorability nouns were drawn from established word memorability norms; verbs were held constant at medium concreteness to reduce noise. Also users are not given only these types of words. A few random words are also introduced. To differentiate the entries, the dataset has an attribute “isTarget” if its true then it means that its one of the 8 categories. Else its some random one.

The experiment was divided into blocks. There were three blocks containing a rest phase in between. Initially Participants were shown practice sentences, then each of the blocks start one after the other.

2.2 Participants

A total of 334 participants were included in the experiment (according to the text file provided with the dataset). But we are performing the analysis on only part of them [232 - 398] which is 114 and we removed some of the participants' data (2 participants). So there are 112. Participants were excluded based on the validation criterion:

- Correct IR > (Wrong IR / 2) + Missed IR (assuming already done before)
- All Blocks that failed this validation test were excluded from memorability score calculation. (assuming that this is already done)

2.3 Variables

Independent Variables:

- Word Memorability Type (HH, HL, LH, LL)
- Voice (Active, Passive)

Dependent Variable:

- Corrected Memorability Score

Corrected memorability was calculated as:

- Corrected Memorability = Hit Rate − False Alarm Rate

Hit Rate is the proportion of repeated target sentences in a condition that the participant correctly identified (Hits ÷ (Hits + Misses)). False Alarm Rate is the proportion of first-presentation sentences on which the participant incorrectly pressed the spacebar (which means the sentence is shown for the first time but participant marks it as seen), calculated globally across all non repeat, non-validation presentations and subtracted uniformly from every condition's hit rate.

2.4 Block Validation and Data Processing :

Each session was divided into three blocks separated by rest breaks. To screen for inattentive participants, each block contained validation sentences (Nearly 10). The dataset has a Validation sentence column where true represents the validation sentence. A block was retained only if the participant passed the following criterion:

Correct Validation IRs > (Wrong Validation IRs ÷ 2) + Missed Validation IRs

Correct = correctly identified, Wrong = Incorrectly identified, Missed = did not respond at all.

Processing steps applied to the raw log data:

- Practice phase rows and timing bookkeeping rows were removed for all the users.
- Sessions were splitted into blocks based on Rest Phase Row.
- The attention-check criteria were applied to each block.
- For each block, if its valid, target-type repeat sentences were classified as Hits or Misses.
- False alarms were counted from non-repeat presentations.
- Corrected memorability was computed per condition. The final analysis dataset contained 896 records (112 participants × 8 conditions), with N = 224 observations per condition.

3. Data Preparation

1. **Loading and cleaning:** Data was given to us as logs. We converted them to CSVs first. Practice rows and gap times were not considered. The flag columns like isTarget, isValidation, and isRepeat were converted to Booleans.
2. **Block segmentation**: Each session was split into three blocks at 'Rest Phase started' marker rows, producing three separate DataFrames per participant.
3. **Attention-check validation**: The criterion formula was applied to each block. Out of 342 total blocks (114 participants × 3 blocks), 7 blocks were excluded as invalid. Two participants had all three blocks fail and were fully removed. So now 112 are left.
4. **Signal detection counting**: For each valid block, target-type repeat sentences (isRepeat = True, isTarget = True, isValidation = False) were identified. We checked up to three subsequent event rows to determine whether an 'IR pressed' event occurred before the next sentence. If yes, the trial = Hit; else = Miss. Thus, False alarms were counted as 'IR pressed' events following non-repeat, non-validation sentences.
5. **Corrected memorability**: Hit rates and the global FA rate were accumulated across all valid blocks per participant. Corrected memorability (Hit Rate − FA Rate) was then computed for each of the 8 conditions. The final dataset contains 896 participant × condition records, aggregated at the participant level.

4. Descriptive Statistics

4.1 Memorability by Word Type

According to Table 1, The HL condition had the highest mean (M = 0.730, Mdn = 0.771), closely followed by HH (M = 0.728, Mdn = 0.771) and LH (M = 0.718, Mdn = 0.757). The LL condition has lowest mean (M = 0.671, Mdn = 0.698), approximately 0.06 units below the top conditions. A difference that is visually apparent in Figure 1a and 1b. All four means are well above zero, confirming that participants reliably recognised repeated sentences far above chance in every condition.

Table 1 : Descriptive statistics of sentences based on word memorability type.

Word Type	N	Mean	Median	SD	IQR	Min	Max
HH (High–High)	224	0.728	0.771	0.200	0.278	−0.094	1.000
HL (High–Low)	224	0.730	0.771	0.192	0.271	−0.035	1.000
LH (Low–High)	224	0.718	0.757	0.200	0.280	−0.069	0.979
LL (Low–Low)	224	0.671	0.698	0.218	0.279	−0.094	1.000

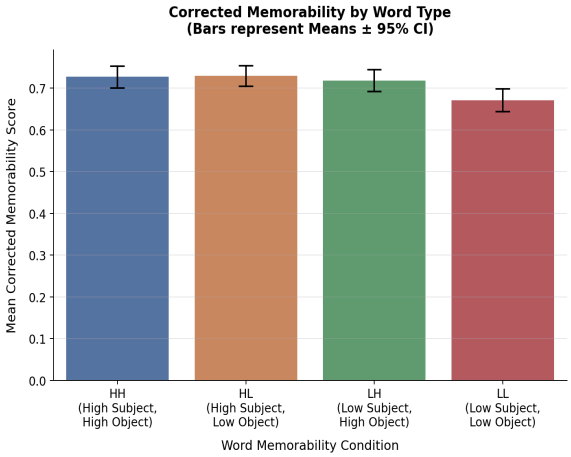

Figure - 1a : shows a bar chart of the mean corrected memorability for each condition, with error bars representing the 95% Confidence Intervals.
Figure - 1b shows a violin plot of corrected memorability by word type with inner box showing the IDR and median

4.2 Memorability by Voice :

Table 2 presents statistics by sentence voice. Active and passive sentences showed nearly identical memorability: means differ by only 0.009 units (Active M = 0.707; Passive M = 0.716) and medians by 0.021 (Active = 0.743; Passive = 0.764). Standard deviations are comparable (≈ 0.20–0.21). Both voice conditions span the same range (approximately −0.09 to 1.00).

Numerically, passive sentences score marginally higher in every word-type condition (Table 3), but the differences are tiny (≤ 0.012) and inconsistent across participants. The SD and IQR are almost identical across active and passive, indicating that voice does not produce systematically more or less variable memory performance.

Table 2. Descriptive statistics for corrected memorability scores by sentence voice.

Voice	N	Mean	Median	SD	IQR	Min	Max
Active	448	0.707	0.743	0.200	0.278	−0.094	1.000
Passive	448	0.716	0.764	0.207	0.292	−0.094	1.000

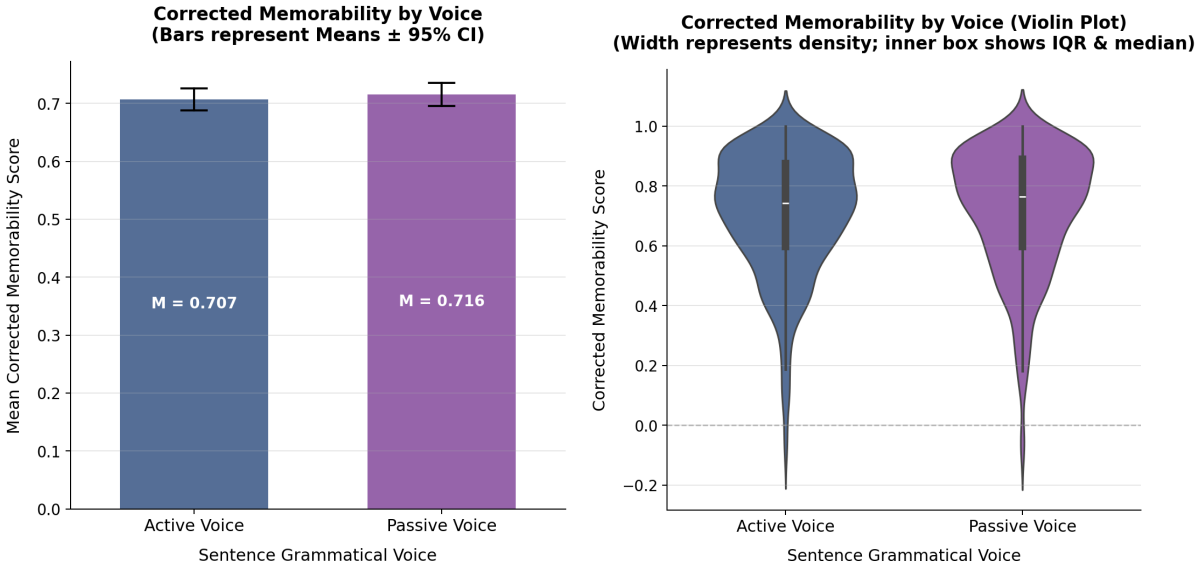

Figure-2a : Bar plot representing Corrected memorability scores by sentence voice along with 95% confidence interval. Figure - 2b : Violin plots representing the Corrected Memorability score with inner box showing IQR and Medians

4.3 Interaction Exploration :

Table 3 presents mean corrected memorability for all eight conditions. Several patterns are worth noting for the interaction between word type and voice. First, passive sentences scored marginally higher than active sentences in every word-type condition — but the gap is consistently small (≤ 0.012) and holds no obvious pattern of widening or narrowing across conditions. Second, and more importantly, the ordering of word types is identical in both voice conditions: HL ≥ HH > LH > LL regardless of voice. This consistent ordering is the clearest early indication of a main effect of word type rather than an interaction.

Table 3. Mean corrected memorability for all 8 conditions, with pooled hit rates and false alarm rates (FA Rate identical across word types because it is computed globally per participant).

Word Type	Active Mean	Passive Mean	Hit Rate	FA Rate
HH (High–High)	0.722	0.734	0.849	0.121
HL (High–Low)	0.728	0.732	0.850	0.121
LH (Low–High)	0.716	0.720	0.839	0.121
LL (Low–Low)	0.665	0.677	0.791	0.121

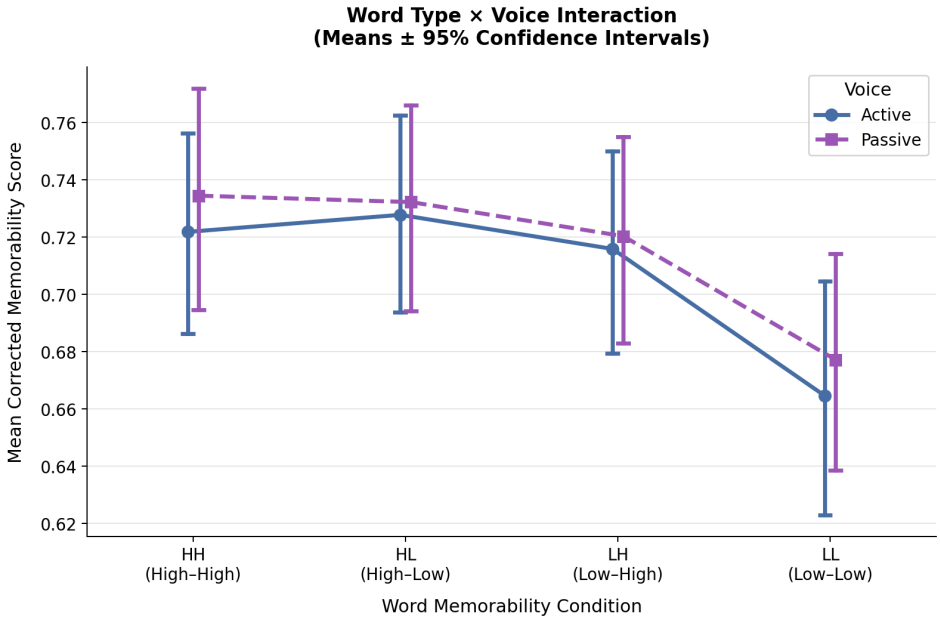

Figure - 3 : Interaction plot: mean corrected memorability by word-type condition and voice. Error bars represent ±1.96 × SEM (approximate 95% CI of the mean). The near-parallel lines and consistent HL ≥ HH > LH > LL ordering across both voice conditions indicate no meaningful interaction between word type and voice.

Does voice affect all word types equally? Based on Figure 3, the lines for Active and Passive run nearly parallel across all four word-type conditions — the gap between the two voice lines are slightly noticeably widen or narrow as word type changes from HH to LL. This suggests that voice does not differentially modulate the memorability advantage conferred by high memorability nouns.
D*oes the gap between HH and LL change across voices?* The HH–LL gap is 0.057 in the Active condition (0.722 − 0.665) and 0.057 in the Passive condition (0.734 − 0.677) essentially identical.

4.4 Normality Assumptions :

Before proceeding with inferential statistics, we tested the assumption of normality using the Shapiro-Wilk test. The corrected memorability scores exhibited a heavily skewed distribution, and the Shapiro-Wilk tests for all conditions were statistically significant ($p < 0.001$), indicating that the data is strictly non-normal. Thus we need non parametric tests for further analysis. We see that from Figure 4, the graph looks quite different from the actual normal distribution. From Shapiro wilk tests and visual graphs, we can conclude that distribution is not following the Normal distribution. So parametric tests cannot be performed.

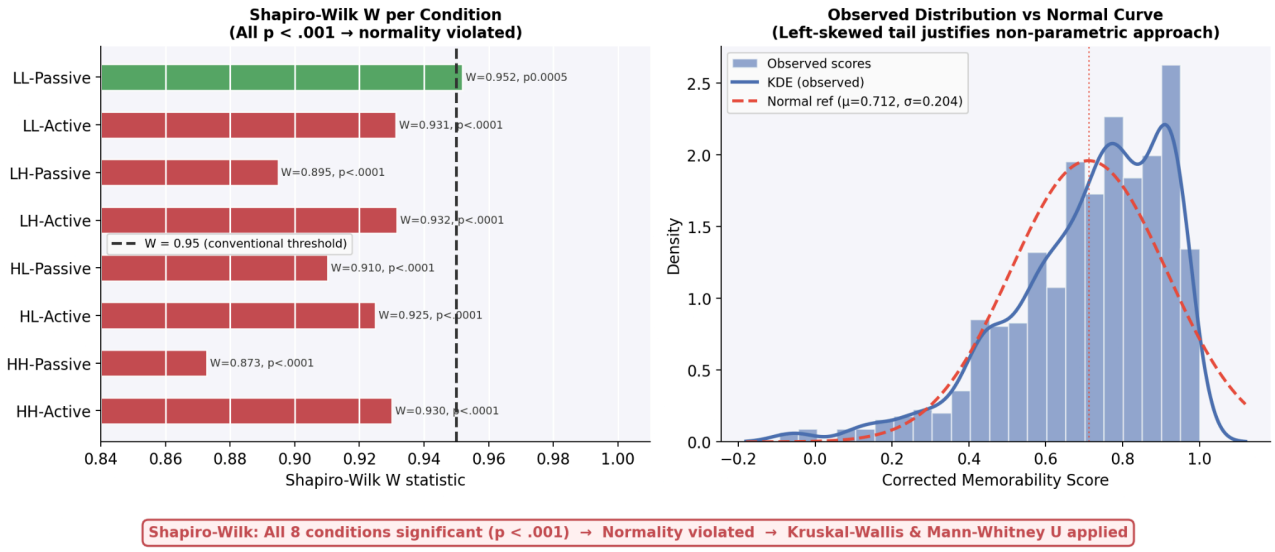

Figure 4 : visualizations of Shapiro-wilk test results and differentiating between expected normal distribution vs actual normal distribution using Shapiro-wilk test.

Final Conclusion

Future Work